

mjablon@iastate.edu

Iowa State University GPA - 3.80
B.S. Computer Engineering - May 2025
M.E. Computer Engineering - May 2026 (Expected)

- **Design Engineer @ GARMIN** May 2025 - August 2025
 - Engineered PCBs for pre-prototype wireless performance testing
 - Root-caused power bugs and reported findings upwards
 - Developed improvements to power & performance in existing technologies
- **Electrical Engineer @ Vermeer** May 2023 - August 2024
 - Quickly built MVP devices for future products
 - Designed electronic circuit and PCBs
 - Analyzed wireless range of prototypes
- **Software Engineer @ RACOM** May 2022 - August 2022

- **Project Director @ PRISUM Solar Car** January 2024 - June 2025
 - Organized and lead a >100 Strong team
 - Large focus on improving management structure
 - Huge progress technically and organizationally
- **Systems Director @ PRISUM Solar Car** July 2022 - January 2024
 - Oversaw technical operations of >50 team members
 - Managed short-term and multi-generational timelines
 - Coordinated inter-team discussions regarding complex vehicle systems
- **Electrical Team Member @ PRISUM Solar Car** August 2021 - July 2022
 - Developed circuit schematics and PCB designs with other members for important subsystems
 - Presented schematics and board designs at formal design reviews
 - Designed BPS control circuitry eliminating hundreds of failure points

- High Voltage Motor Controller for lightweight vehicles
 - Application expert for our Senior design team
 - Continued design independently to further optimize efficiency and size
 - Current prototype less than ½ the size of competing solutions
 - GaNFet Drivers and Field-Oriented-Control
- Built basic CPU on Breadboards from elementary logic circuits
 - Designed architecture, layout, microcode, machine code language, and basic compiler
- Designer of prototype battery monitoring system
 - Consolidates current modular designs into a more efficient form factor while not reducing the customizability and workability of the current system
- Creator of an annual Animated-to-music Christmas Light show
 - System design, construction, sequencing

- Ability to quickly understand high-level systems in detail
- Electrical Circuit and PCB Design
- Reverse-Engineering, Troubleshooting
- Linux environment, User Interface (UX), Virtual Machines, CAD, Computer Vision
- Building electromechanical systems to accomplish complex tasks

- Altium Designer, Kicad, Cadence System Capture, Allegro, LTspice
- Autodesk Inventor, Fusion 360, Solidworks, ANSYS
- C, Java, JavaScript, PHP, Python, HTML, CSS, Verilog, VHDL